

Analytical and Computational Analysis of Hyperbolic Interference Fringes in 3D

In some places like Desmos, to represent λ or wavelength, we have used "l".

Abstract

This paper presents a rigorous investigation into the three-dimensional geometry of interference fringes produced by two coherent point sources. While traditional pedagogical treatments of Young's Double-Slit Experiment (YDSE) often rely on far-field planar approximations, we provide a complete analytical derivation demonstrating that the loci of constant phase difference form a family of hyperboloids of two sheets. This analytical model is complemented by a computational simulation that captures both instantaneous wave superposition and time-averaged intensity. Our findings confirm the core thesis: although the instantaneous intensity field exhibits high-frequency temporal fluctuations, the macroscopic, observable fringe pattern in a laboratory setting is the result of the time-averaged intensity distribution, which aligns perfectly with the derived hyperbolic geometry.

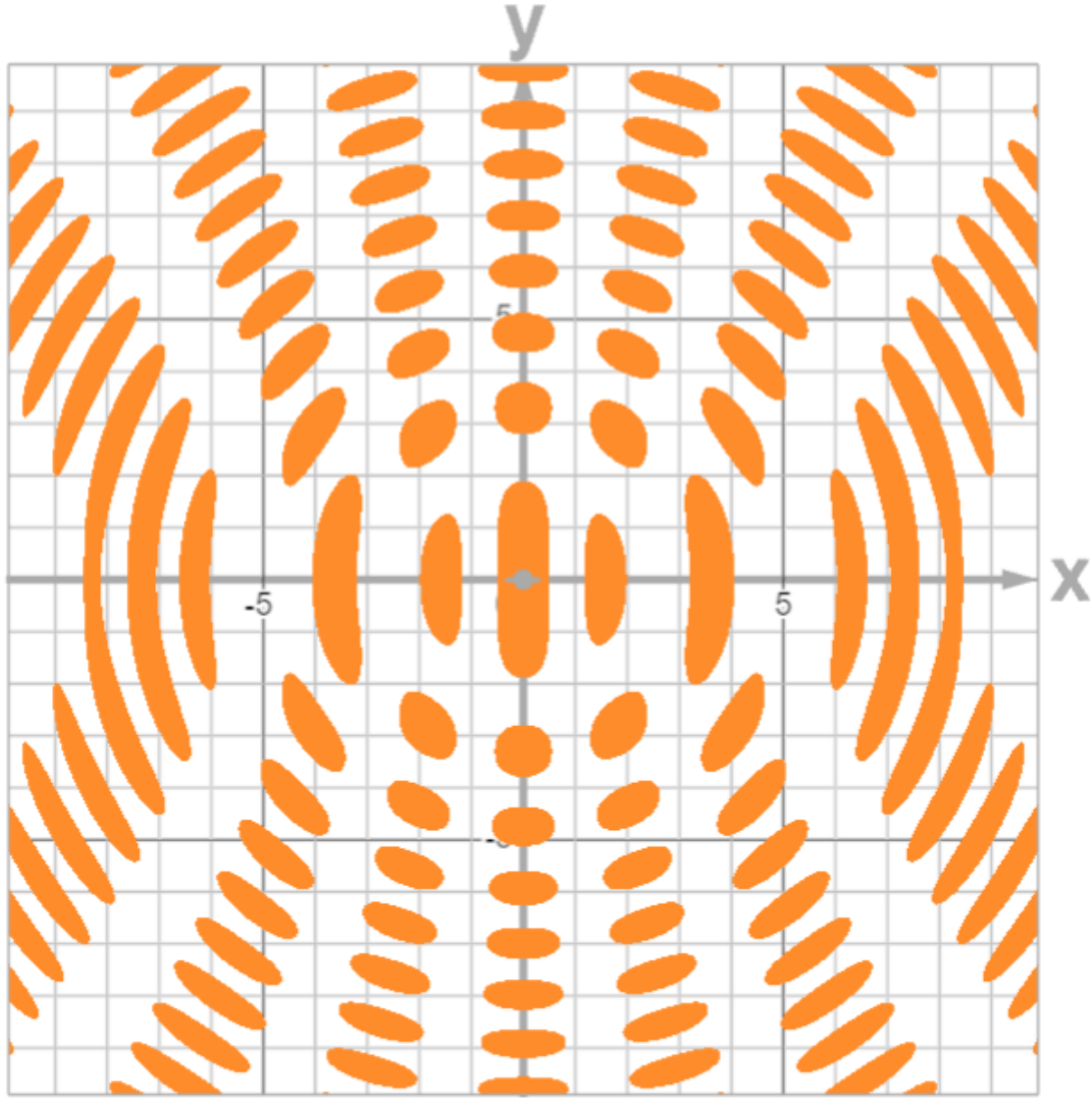


Figure 1: A two-dimensional cross-section in the xy -plane illustrating the loci of constructive interference. The orange regions represent points where the path difference is an integer multiple of the wavelength, forming the characteristic hyperbolic branches that extend into three-dimensional space as hyperboloids.

1. Introduction

Young's Double-Slit Experiment (YDSE) remains a cornerstone of wave optics, providing fundamental evidence for the principle of superposition. In standard 2D approximations, the interference fringes observed on a distant screen are often described as straight, parallel lines. However, a precise 3D treatment reveals a more complex reality. By considering the path difference from two point sources to any point in space, it becomes evident that the surfaces of constructive and destructive interference are curved. This paper bridges the gap between theoretical geometry and computational observation, emphasizing the transition from rapidly oscillating instantaneous fields to the stable, time-averaged intensity maps perceived by detectors and the human eye.

2. Analytical Derivation of the Hyperbolic Fringe Pattern

To determine the geometry of the interference fringes, we define two coherent point sources located on the x-axis at $(d/2, 0, 0)$ and $(-d/2, 0, 0)$. For an arbitrary point $P(x, y, z)$ in space, the distances from the sources are given by:

$$r_1 = \sqrt{\left(x - \frac{d}{2}\right)^2 + y^2 + z^2}$$

$$r_2 = \sqrt{\left(x + \frac{d}{2}\right)^2 + y^2 + z^2}$$

Let $R^2 = y^2 + z^2$ represent the transverse radial distance squared:

$$r_1 = \sqrt{\left(x - \frac{d}{2}\right)^2 + R^2}$$

$$r_2 = \sqrt{\left(x + \frac{d}{2}\right)^2 + R^2}$$

For constructive interference, the path difference Δp must be an integer multiple of the wavelength λ :

$$\Delta p = m \lambda = r_1 - r_2$$

Rearranging and squaring both sides:

$$\begin{aligned} \Rightarrow r_1 &= r_2 + m \lambda \\ \Rightarrow r_1^2 &= (r_2 + m \lambda)^2 \\ \Rightarrow r_1^2 &= r_2^2 + (m \lambda)^2 + 2 r_2 (m \lambda) \\ \Rightarrow r_1^2 - r_2^2 &= (m \lambda)^2 + 2 (m \lambda) r_2 \end{aligned}$$

Substituting the expressions for r_1 and r_2 :

$$\begin{aligned} \Rightarrow \left[\left(x - \frac{d}{2}\right)^2 + R^2 \right] - \left[\left(x + \frac{d}{2}\right)^2 + R^2 \right] &= (m \lambda)^2 + 2 (m \lambda) r_2 \\ \Rightarrow \left(x^2 - x d + \frac{d^2}{4} \right) - \left(x^2 + x d + \frac{d^2}{4} \right) &= (m \lambda)^2 + 2 (m \lambda) r_2 \\ \Rightarrow -2 x d &= (m \lambda)^2 + 2 (m \lambda) r_2 \end{aligned}$$

Isolating r_2 and squaring again:

$$\begin{aligned} \Rightarrow -[2 x d + (m \lambda)^2] &= 2 (m \lambda) r_2 \\ \Rightarrow [2 x d + (m \lambda)^2]^2 &= 4 (m \lambda)^2 r_2^2 \end{aligned}$$

$$\Rightarrow 4x^2d^2 + (m\lambda)^4 + 4xd(m\lambda)^2 = 4(m\lambda)^2 \left[\left(x + \frac{d}{2} \right)^2 + R^2 \right]$$

$$\Rightarrow 4x^2d^2 + (m\lambda)^4 + 4xd(m\lambda)^2 = 4(m\lambda)^2 \left[x^2 + xd + \frac{d^2}{4} + R^2 \right]$$

$$\Rightarrow 4x^2d^2 + (m\lambda)^4 + 4xd(m\lambda)^2 = 4(m\lambda)^2x^2 + 4xd(m\lambda)^2 + (m\lambda)^2d^2 + 4(m\lambda)^2R^2$$

Canceling the cross-terms and grouping variables:

$$\Rightarrow 4x^2d^2 + (m\lambda)^4 = 4(m\lambda)^2x^2 + (m\lambda)^2d^2 + 4(m\lambda)^2R^2$$

$$\Rightarrow 4x^2(d^2 - (m\lambda)^2) - 4(m\lambda)^2R^2 = (m\lambda)^2(d^2 - (m\lambda)^2)$$

Dividing by $(m\lambda)^2[d^2 - (m\lambda)^2]$ to reach standard form:

$$\Rightarrow \frac{4x^2}{(m\lambda)^2} - \frac{4R^2}{d^2 - (m\lambda)^2} = 1$$

$$\Rightarrow \frac{x^2}{\left(\frac{m\lambda}{2}\right)^2} - \frac{y^2 + z^2}{\left(\frac{d}{2}\right)^2 - \left(\frac{m\lambda}{2}\right)^2} = 1$$

This result is the standard equation for a hyperboloid of two sheets, where the fringes are the intersections of these surfaces with the observation screen.

3. Computational Methodology

To validate the analytical derivation and explore the temporal dynamics of the interference, a computational model was developed using Python. The simulation defines two point sources and calculates the resultant wave amplitude at every point on a coordinate grid. The instantaneous intensity I is modeled as the square of the sum of two sinusoidal wave functions:

$$I(t) = (a \sin(\omega t - k r_1) + a \sin(\omega t - k r_2))^2$$

Where a is the amplitude, ω is the angular frequency, k is the wave number, and r_1, r_2 are the distances from the sources. The code iterates over a time array to visualize the high-frequency fluctuations of the wave field. Finally, the time-averaged intensity is computed by integrating the instantaneous intensity over a full period, which is then visualized using [matplotlib.pyplot.contourf](#) to produce the recognizable static fringe pattern.

Young's Double-Slit Experiment

3D Wave Interference Pattern

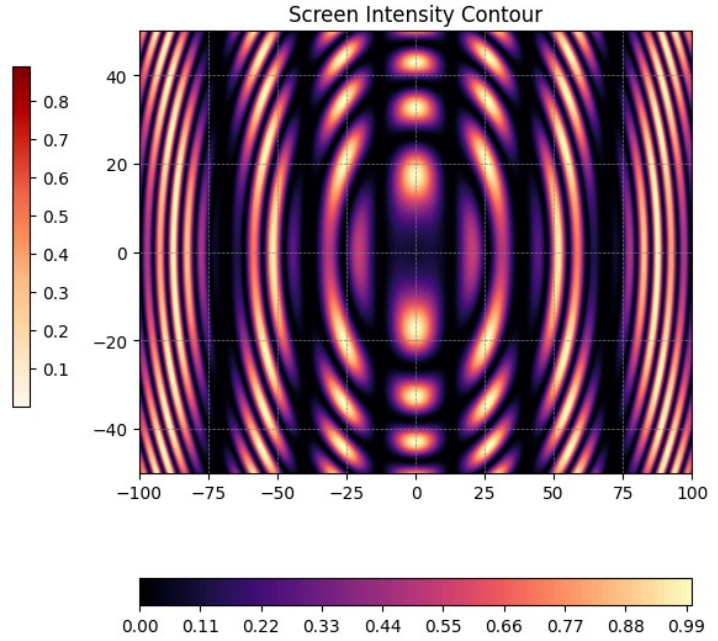
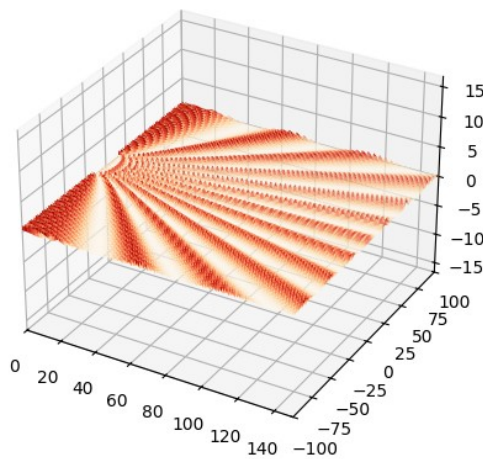


Figure 1: A dual-pane visualization showing the 3D wave interference pattern (left) and the corresponding instantaneous screen intensity contour (right). The snapshot captures the rapid spatial oscillations of the wave field at a single point in time.

4. Discussion: Instantaneous vs. Time-Averaged Intensity

The core of this analysis lies in the distinction between the mathematical wave function and the physical observation. As shown in the computational snapshots, the instantaneous intensity field is highly dynamic, with "hot spots" of constructive interference that shift as the phases evolve.

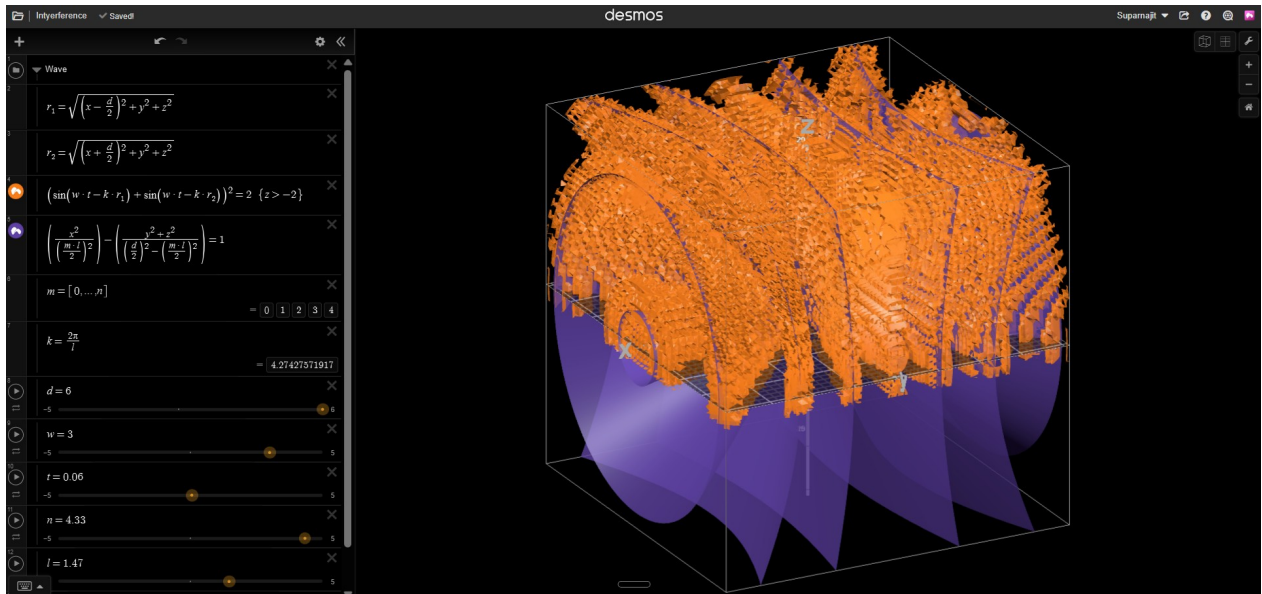


Figure 2: Desmos 3D plot illustrating the surfaces of constant phase difference. The orange regions represent the volume where constructive interference occurs, clearly following the hyperbolic geometry derived in Section 2.

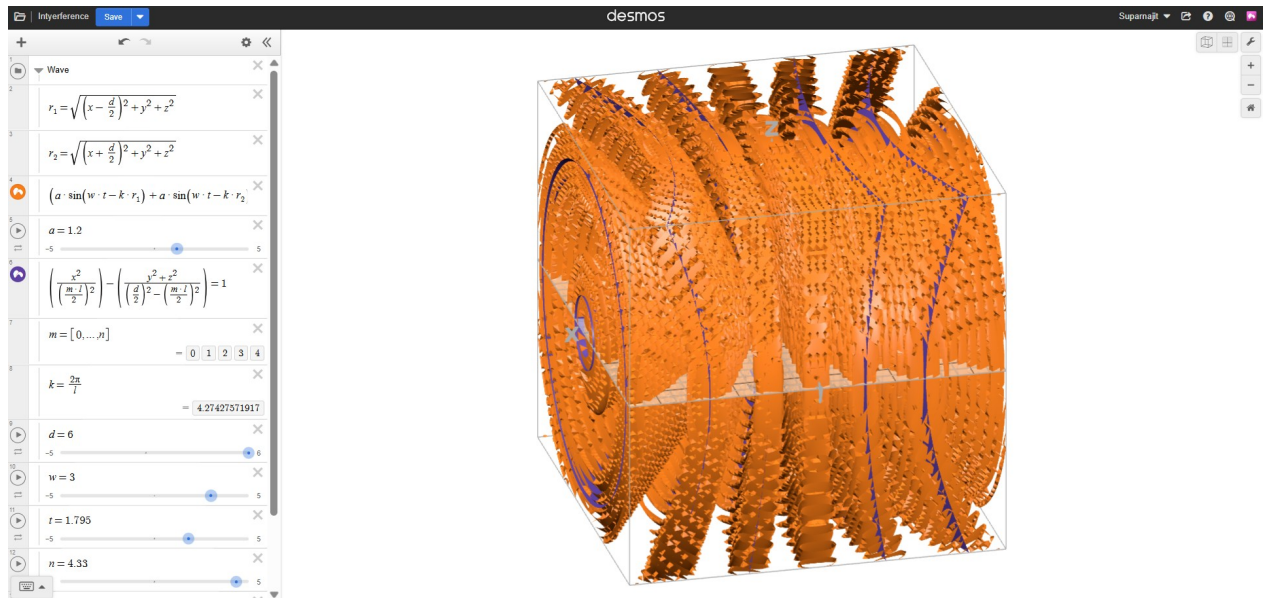


Figure 3: A closer view of the hyperbolic surfaces in Desmos 3D, showing the nested structure of the hyperboloids of two sheets corresponding to different orders of interference (m).

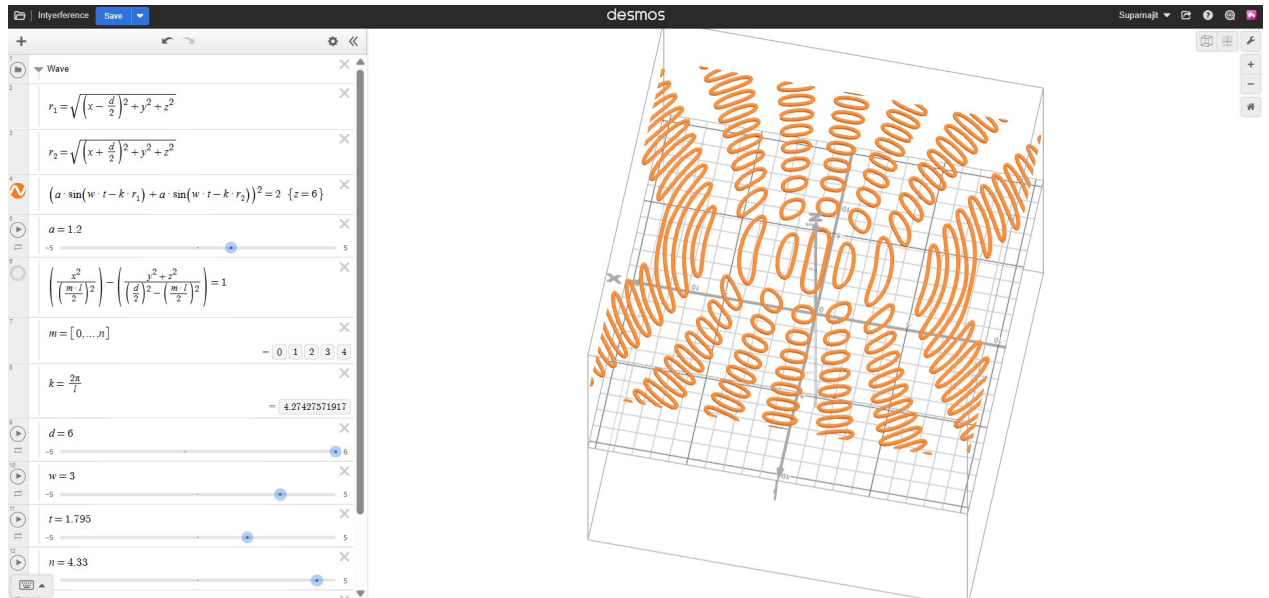


Figure 4: Intersection of the hyperbolic surfaces with a transverse plane ($z = 6$), demonstrating how the 3D hyperboloids manifest as curved fringes on a 2D observation screen.

However, optical frequencies are on the order of 10^{14} Hz, far exceeding the temporal resolution of any physical detector. What is actually observed is the time-averaged intensity:

$$\langle I \rangle = \frac{1}{T} \int_0^T I(t) dt$$

This averaging process smooths out the temporal fluctuations, leaving behind the stable, hyperbolic fringe pattern predicted by the path-difference derivation. The computational results for time-averaged intensity (Figure 5) align perfectly with the theoretical hyperboloid intersections.

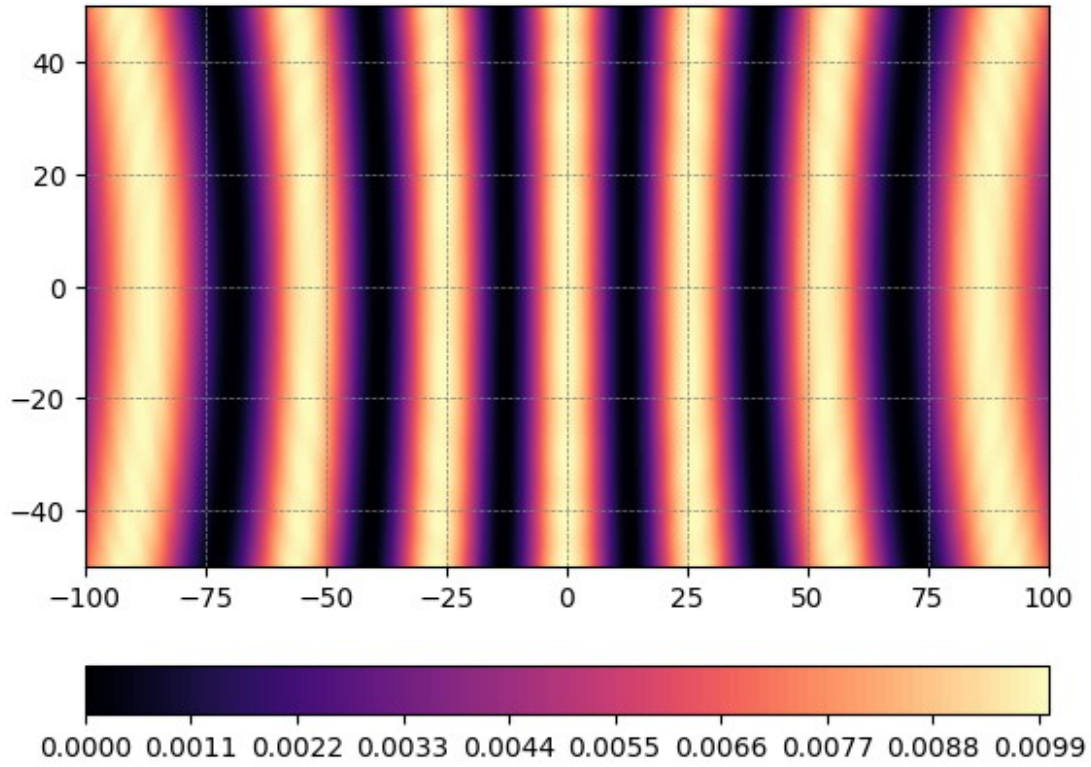


Figure 5: Time-averaged intensity map on the observation screen. The rapid fluctuations seen in Figure 1 are integrated out, revealing the steady-state hyperbolic fringes that characterize the macroscopic interference pattern.

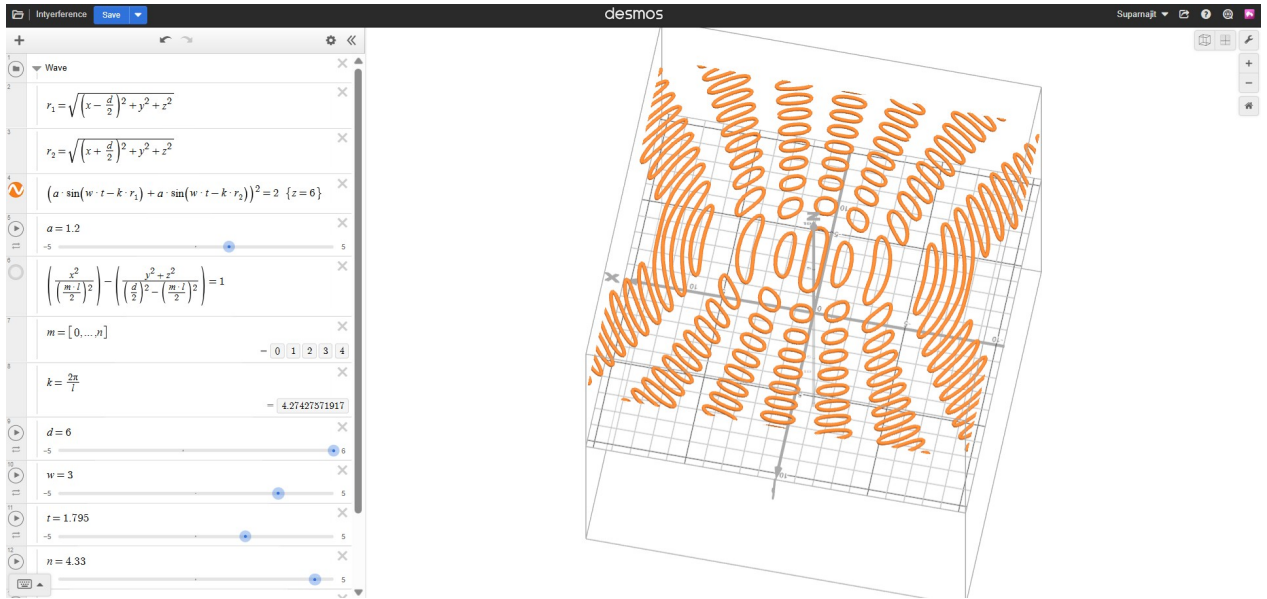


Figure 6: Comprehensive 3D visualization of the interference field and its 2D screen projection, synthesizing the relationship between the volumetric wave interaction and the observed planar intensity.

5. Conclusion

Through a combination of rigorous analytical derivation and computational modeling, we have demonstrated that the interference fringes in a three-dimensional setup are defined by hyperboloids of two sheets. The analytical proof provides the exact geometric framework, while

the Python-based simulation illustrates the transition from high-frequency instantaneous wave superposition to the stable time-averaged intensity patterns observed in reality. This study reinforces the importance of considering full 3D geometries and temporal averaging in the analysis of optical interference, providing a comprehensive understanding of the physical mechanisms behind hyperbolic fringe patterns.